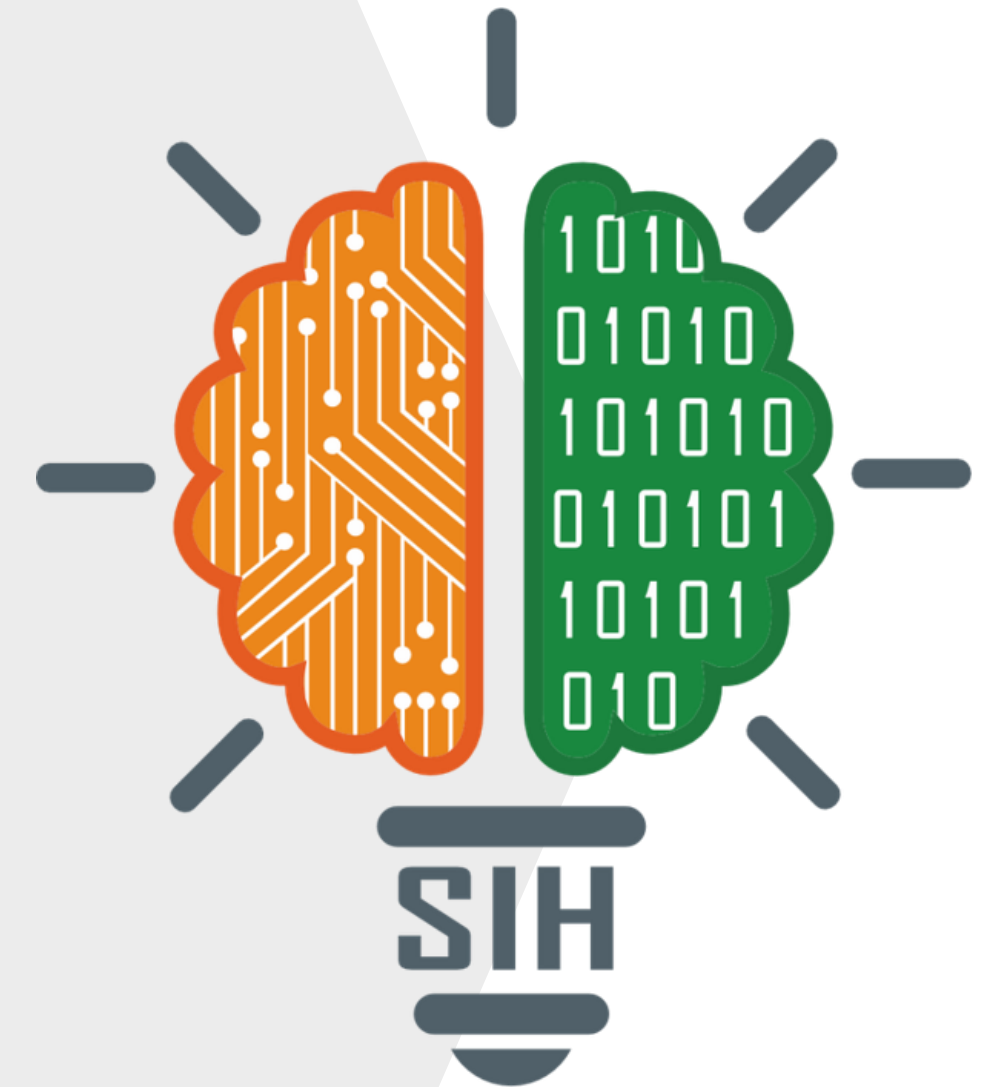


TITLE PAGE

- Problem Statement ID – **SIH26051**
- Problem Statement Title- **Software Based Model Development for Design of Area Specific Shelter for Thermal Comfort Maintenance.**
- Theme- **Miscellaneous**
- PS Category- **Software**
- Team ID-
- Team Name (Registered on portal) - **SHVAAM**





TERRA-SHIELD



REAL LIFE ISSUES



Fuel-based heaters (Bukhari) cause **CO poisoning** and fatalities at high-altitude posts.



Existing solutions are standalone hardware, built by trial and error.



No software to simulate or validate designs before physical deployment.



WHY IMPORTANT



DRDO's redesign cut fuel consumption by ~50% (saves thousands of crores).



Fuel convoys to remote posts are costly and high-risk.



Better passive shelter design protects lives, reduces fuel dependency, and enables faster, transparent decision-making.



**LESS FUEL
SAFER SOLDIERS
STRONGER INDIA**



SOLUTION



Web platform: input location, geometry, materials, orientation.



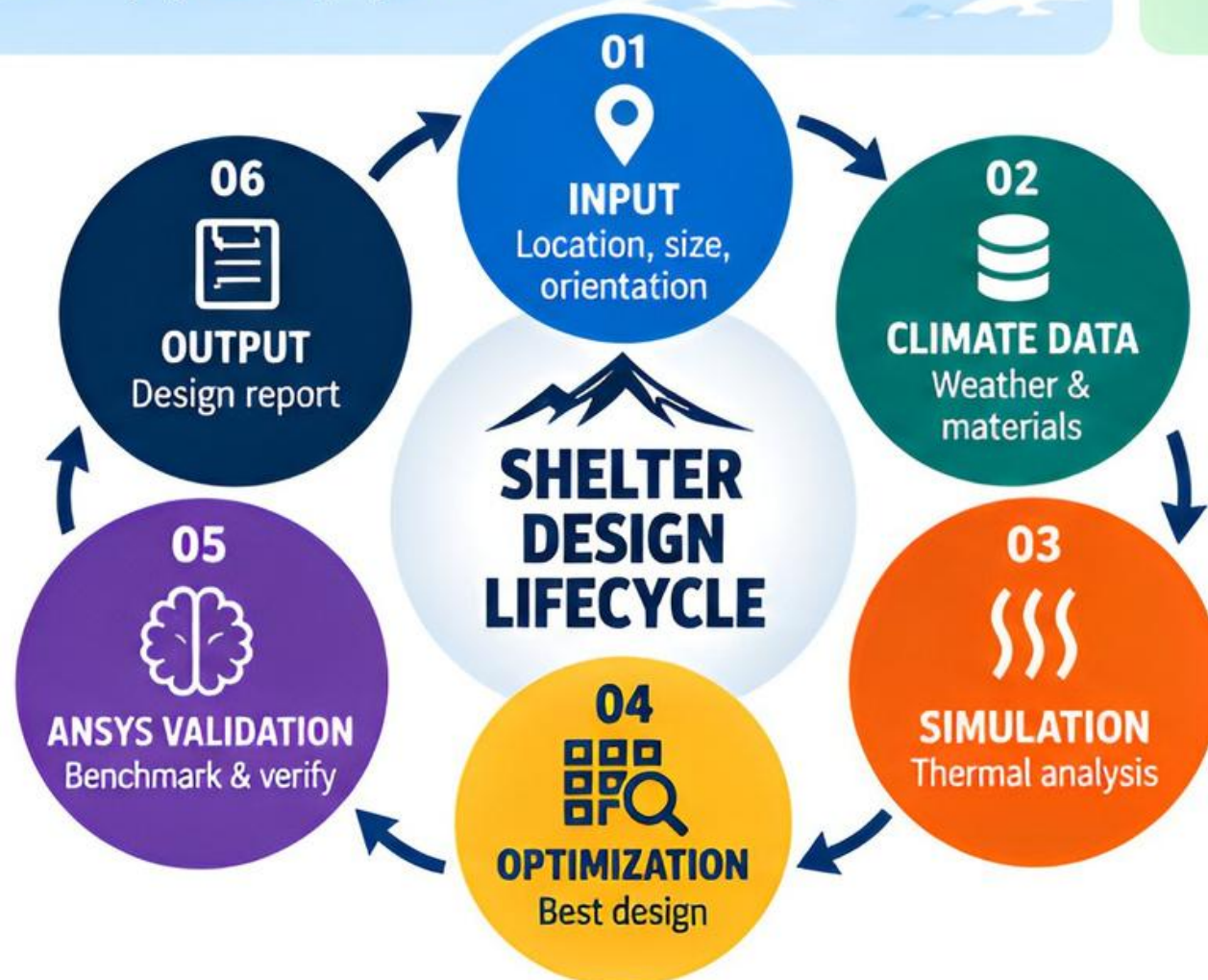
Physics-based thermal simulation (conduction, convection, radiation, solar gain) gives instant results.



Optimization engine tests thousands of configurations and returns the best design for comfort, fuel, weight and cost (validated with ANSYS).



**SIMULATE
OPTIMIZE
DEPLOY BETTER**



VALIDATED

ANSYS benchmarked (low error)



OPTIMIZED

Best design for comfort, fuel, weight, cost



TERRAIN-SPECIFIC

Presets for Siachen/Ladakh, desert and North-East





TECHNICAL APPROACH



TECH STACK



PHYSICS / OPTIMIZATION

NumPy, SciPy, Pandas,
custom transient
heat-transfer model



WEB APP

React + Vite, TypeScript,
Plotly for graphs



BACKEND

Python, FastAPI,
PostgreSQL
material/climate database



VALIDATION & DATA

Offline ANSYS benchmark
runs, NASA POWER climate
API, cached datasets for
offline demo reliability



THERMAL SIMULATION & OPTIMIZATION PIPELINE



1 CLIMATE + MATERIAL INPUT

Location, weather data
(NASA POWER),
material properties



2 PARAMETER PROCESSING

Pre-processing,
boundary conditions,
thermo-physical
properties



3 TRANSIENT THERMAL MODEL

Conduction,
convection, radiation,
solar gain



4 GRID-SEARCH OPTIMIZATION

Thousands of
configurations
ranked



5 RANKED DESIGN OUTPUT

Best design
returned

Offline ANSYS cross-check on benchmark configs



SHELTER DESIGN PROCESSING PIPELINE

1



INPUT SHELTER SPEC

Geometry, dimensions,
material options,
constraints

2



TERRAIN PRESET MATCH

Auto-applies Ladakh /
desert / North-East /
and profile

3



COMFORT SCORE GATE

Checks if design meets
target indoor conditions

4



PARETO OUTPUT & EXPORT

Generates ranked CSV +
comparison charts



TERRAIN-PRESET & RETROFIT WORKFLOW

1



SELECT / DETECT TERRAIN PRESET

Auto-loads cold-desert / arid /
humid-NE climate + constraint profile

2



RUN NEW-BUILD OR RETROFIT MODE

Secure local computation,
no internet dependency
for cached climate data

3



REVIEW FLAGGED TRADE-OFFS

Borderline configs routed to
manual engineer review rather
than auto-accepted



DIGITAL OUTPUT



Ranked Design
Configuration Report



Comfort vs Fuel vs Weight
vs Cost Pareto Chart



ANSYS Cross-Validation
Summary



Terrain Preset
Justification Sheet



Retrofit Recommendation
List (for existing structures)



PHYSICS-VERIFIED

Every recommendation traceable
to explicit heat-transfer equations
and grid-search scoring —
no black-box AI claim.



FEASIBILITY AND VIABILITY

PRACTICAL • IMPACTFUL • SCALABLE



SMART INDIA
HACKATHON
2026



FEASIBILITY

REAL TO BUILD
READY TO DEMO



No Specialized Infrastructure

Standard stack (Python/FastAPI, React, NumPy/SciPy); no proprietary hardware, no lab equipment needed.



Physics, Not Exotic ML

Conduction/convection/radiation/solar-gain equations are standard heat-transfer theory; optimization is grid search, fully explainable to judges.



De-Risked ANSYS Dependency

ANSYS run offline on benchmark configurations pre-event, not live during demo, removing the single largest point of failure.



Modular Rollout

Start with one terrain preset (Ladakh cold-desert) before expanding to desert and North-East presets.



Realistic Team Skill Ask

Python + basic frontend + one member comfortable with heat-transfer equations; runs entirely on a laptop.



BUSINESS POTENTIAL

FROM SOLUTION
TO REAL-WORLD
IMPACT

01



Government Adoption — Priority Pilot

B2G deployment starting with a single DRDO/Army test post.

02



Institutional Licensing

Per-post or per-unit licensing model for defence deployment.

03



Dual-Sector Platform

Same engine repackaged as a homestay/guesthouse thermal-design advisory tool for Ladakh's growing tourism-construction market.

04



Retrofit Advisory Service

Standalone paid service ranking minimum-cost upgrades for existing shelters/huts.



VIABILITY

ALIGNED TODAY
SUSTAINABLE
TOMORROW



Direct Operational Fit

Mirrors DRDO's existing move toward passive-solar, low-fuel shelters (Him Tapak, DIHAR solar systems), not a new unproven direction.



Institutional Value

Gives defence planners a design/decision layer before physical builds, replacing trial-and-error hardware iteration.



Cost & Risk Reduction

A wrong physical shelter prototype at altitude costs real money and logistics; simulate-first avoids that expense before committing to a build.



Fast-Moving Civilian Track

Ladakh's tourism boom and government "industry status" incentives for solar-passive homestay construction create an immediate, self-funding private-sector user base.



Scalable Architecture

Same core engine extends to new terrain presets and shelter types without rebuilding the system.



IMPACT AND BENEFITS

SAFER SHELTERS • STRONGER FORCES • A MORE RESILIENT INDIA



POTENTIAL IMPACT ON THE TARGET AUDIENCE



SOLDIERS & FIELD PERSONNEL

- **Safer shelters** that eliminate reliance on backblast-prone fuel heaters at extreme altitude.
- Designs validated digitally before deployment, reducing risk of field failure in -40°C conditions.
- Faster access to comfort-optimized shelter configs suited to their exact post location.



DEFENCE ENGINEERS & PLANNERS

- Rapid comparison of material/orientation/insulation options without physical prototyping.
- ANSYS-benchmarked confidence in outputs without running full CFD every time.
- Clear ranked recommendations instead of manual trial-and-error iteration.



DRDO / MINISTRY OF DEFENCE

- Faster, cheaper design iteration cycles across multiple terrain types and post locations.
- Digital record of design rationale — reproducible, defensible, and auditable.
- A reusable design layer that could inform future hardware programs like Him Tapak faster.



SOCIAL BENEFITS

PEOPLE
SAFER, STRONGER, TOGETHER



Improved Soldier Safety — reduces carbon-monoxide and backblast risk from legacy heaters.



Faster Design-to-Deployment — digital simulation shortens the trial-and-error cycle before a shelter reaches the field.



ECONOMIC BENEFITS

COST EFFICIENT
LONGER LASTING VALUE



Reduced Prototype Cost — avoids expensive physical iterations at high-altitude logistics cost.



Reusable Platform Value — same engine supports future terrain types and shelter programs without rebuilding from scratch.



Civilian Revenue Stream — repackaged as a homestay/guesthouse thermal-design advisory tool for Ladakh's tourism-construction boom.



REGULATORY & TECHNICAL BENEFITS

TRUSTED BY STANDARDS
BUILT FOR REAL CONDITIONS



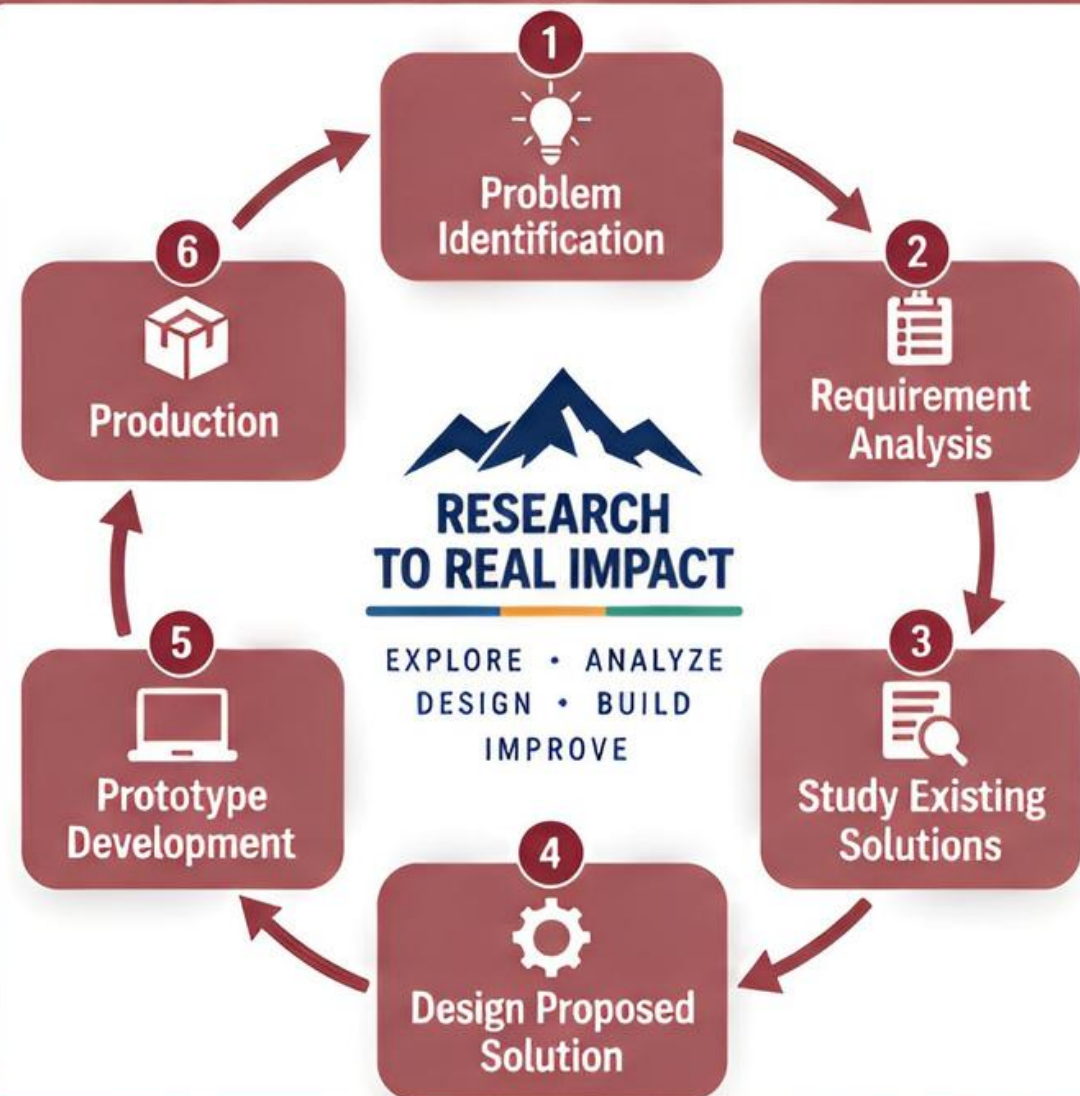
ANSYS-Aligned Credibility — offline-benchmarked physics engine matches the PS's explicit ANSYS reference without a fragile live dependency.



Terrain-Specific Compliance — built-in presets for cold-desert, arid border, and humid North-East conditions match DRDO's actual operational geography.



RESEARCH WORKFLOW



RESEARCH GAPS AND OUR SOLUTION

CAPABILITY	EXISTING SIMULATION TOOLS (EnergyPlus / IES VE / Honeybee)	MILITARY FIELD PRACTICE TODAY (manual/trial-and-error)	THERMASHIELD
High-altitude / defence terrain presets	✗ Generic building climates only	✗ No structured presets, field experience only	✓ Built-in Siachen/Ladakh, arid, humid North-East presets
Multi-objective trade-off (fuel/weight/cost)	✗ Energy-cost focus only, no transport-weight dimension	✗ No formal trade-off analysis	✓ Pareto frontier: comfort vs fuel vs weight vs cost
ANSYS-referenced validation	✓ Some tools support CFD linkage	✗ None	✓ Offline ANSYS-benchmarked physics engine
Rapid design-space search	✓ Parametric analysis tools exist	✗ Manual, slow, one design at a time	✓ Automated grid search across thousands of configs
Retrofit / upgrade advisory for existing structures	✗ Not a built-in workflow	✗ No systematic method	✓ Ranked minimum-cost intervention list



REFERENCES

1. DRDO Him Tapak (official product page) — <https://drdo.gov.in/space-heating-device-bukhari-himtapak>
2. DRDO DIHAR — solar thermal heating tech transfer (Leh) — <https://www.energetica-india.net/news/solarium-signs-technology-transfer-agreement-with-drdo-for-solar-thermal-heating>
3. DRDO Defence Laboratory Jodhpur — Phase Change Materials for Defence Applications — <https://publications.drdo.gov.in/ojs/index.php/dsj/article/download/363/557>

4. G.B. Pant National Institute of Himalayan Environment — Practical Guide for Passive Solar Heated Buildings — https://nmhs.org.in/w/pdf/Key_Publications/PSHBs.pdf
5. Hosamo et al. (2022) — BIM + Machine Learning + NSGA-II Multi-Objective Optimization of Building Energy & Thermal Comfort — https://vbn.aau.dk/ws/portalfiles/portal/502995517/1_s2.0_S0378778822006508_main.pdf